RESEARCH PARADIGM: DESIGN SCIENCE APPLIED TO DIGITAL TWINS IN CONSTRUCTION

Doctoral Candidate: Jhon Wilbert Quiroz Quilla

Course: Research Methods and Scientific Integrity in AI and Advanced Technologies

Program: Doctoral Program in Deep Technologies – UNMSM

1. RESEARCH TOPIC AND CONTEXT

Project Title

"Intelligent Digital Twin Model with Cognitive Learning for Prediction and Optimization of Costs, Schedules, and Risks in Construction Projects"

Problem Context

The construction industry in Peru faces a silent but devastating crisis. According to the latest report from the Peruvian Chamber of Construction (CAPECO, 2025), 87% of building projects in Metropolitan Lima present deviations greater than 15% from their original budget, and 76% exceed contractual deadlines. Globally, the situation is equally alarming: 98% of infrastructure projects suffer cost overruns or delays, with an average cost overrun of 80% above the original budget (Flyvbjerg, 2024).

I have had the opportunity to speak with project managers from at least five medium-sized construction companies in Lima, and they all agree on a critical point: the lessons learned from past projects—specifically regarding actual versus planned costs and activity durations—tend to remain stored in disconnected files. This systematic loss of digitalized "institutional memory" prevents companies from learning from their past mistakes to anticipate future risks.

The question that has guided me since the beginning of this research is: how can we build a system that not only digitally represents a construction project, but also learns from accumulated experience to predict and optimize costs, schedules, and risks?

My Personal Motivation

I have worked in the construction sector for over a decade, and I have seen how promising projects become financial nightmares simply because we could not anticipate what had already happened before. I have seen site supervisors with decades of experience make decisions based on their intuition, and I have seen how that intuition, although valuable, does not always translate into optimal decisions. I have seen how historical data—which could have prevented errors—gets lost in forgotten Excel files or in the memory of professionals who leave the company.

This research is born from that frustration and that hope: the hope that technology can help us not repeat the same mistakes, that an intelligent system can learn from our past to build a better future. This is not about replacing human judgment, but about augmenting it with the memory that we, as an industry, have been unable to preserve.

2. PRELIMINARY RESEARCH QUESTION

Main Research Question

How can an intelligent digital twin model with cognitive learning, using historical project data, contribute to the prediction and optimization of costs, schedules, and risks in construction projects in the Peruvian context?

Sub-Questions

1. How to design a digital twin architecture that integrates historical project data, BIM data, and contextual variables to faithfully replicate the dynamic behavior of a construction project?

2. What cognitive learning techniques (machine learning, deep learning, case-based reasoning) improve the prediction of costs, schedules, and risks in construction projects with limited data volumes?

3. How to structure a digital twin model that enables "what-if" scenario simulation and real-time decision optimization?

4. To what extent does the implementation of the proposed model reduce budgetary, schedule, and risk deviations compared to traditional project management methods?

Why This Question Matters

This question is important because it addresses a problem that affects not only construction companies, but Peruvian society as a whole. Cost overruns and delays in construction projects mean fewer hospitals, fewer schools, fewer roads. They mean that public resources are wasted instead of invested in the country's development. They mean that Peruvian companies lose competitiveness against those that have adopted predictive technologies. Ultimately, this question matters because construction is not just an economic sector; it is the materialization of our collective dreams as a society.

3. SELECTED PARADIGM AND JUSTIFICATION

Selected Paradigm: Design Science

The paradigm guiding this research is Design Science, an approach that, as Hevner and colleagues (2004) point out, seeks to "extend the boundaries of human and organizational capabilities by creating new and innovative artifacts." Simon (1996), Nobel laureate and pioneer of this paradigm, defined it as "the attempts to create things that serve human purposes."

In the context of my research, the central artifact is the intelligent digital twin model with cognitive learning, specifically designed for the context of the Peruvian medium-sized construction company.

Justification of the Choice

Why design science and not another paradigm?

I have reflected deeply on this decision, and I want to share my thought process. Initially, when I began formulating this research, my instinct was to adopt a positivist approach. After all, I was working with numbers: costs, schedules, performance metrics. I felt comfortable with numbers. But as I delved deeper into the problem, I realized that positivism alone was not enough.

Rejection of pure positivism: Measuring cost overrun rates or delay days is necessary, but it is not sufficient. Positivism would tell me how much we are deviating, but it would not help me build a tool to prevent those deviations. A purely positivist approach would give me diagnoses, but not solutions. And I do not want to just diagnose the problem; I want to help solve it.

Rejection of pure interpretivism: An interpretivist approach would allow me to understand why project managers make certain decisions, why they trust their intuition more than data, why lessons learned are not documented. But interpretivism alone would not give me a tangible artifact that companies can use. Understanding the problem is important, but it is not enough if we do not build something that solves it.

Rejection of pure mixed methods: Mixed methods combine quantitative and qualitative approaches, which is valuable. But in my case, the emphasis is not on describing or explaining a phenomenon, but on building a solution. Mixed methods would help me understand the problem, but the design science paradigm allows me to go one step further: it allows me to create.

Why is design science the right choice?

Because I am building an artifact: The digital twin model with cognitive learning is a tangible artifact that solves a practical problem. It is not just a theory or an explanatory model; it is a tool that construction companies can implement.

Because the contribution is utility: The success of my research is measured not only by hypothesis confirmation, but by the utility of the artifact. Does it reduce cost overrun prediction error? Is it accessible to Peruvian medium-sized companies? Can professionals use it without needing to be artificial intelligence experts?

Because knowledge emerges from construction: As Hevner et al. (2004) point out, "knowledge and understanding of a problem domain and its solution are achieved in the construction and application of the designed artifact." It is not just about observing the problem, but about building it, testing it, failing, and learning in the process.

Because it addresses a "wicked" problem: Construction project management problems are "wicked" in the sense that they have unstable requirements, complex interactions, and depend critically on human capabilities. Design science is made for this type of problem.

Reflection on My Choice

I confess that this choice was not easy. For weeks, I wavered between positivism and design science. My engineering background inclined me toward numbers, toward objective measurements. But something did not fit. If I only wanted to measure cost overruns, what was I contributing that a financial report could not do? My added value is not in measuring the problem, but in building the solution.

This decision has made me more aware of my own bias. As an engineer, I tend to think about technical solutions before fully understanding the problem. Design science forces me to balance both things: to build, but also to evaluate; to create, but also to justify. It forces me to be humble, to acknowledge that my artifact may not work, and to document both failures and successes.

4. IMPLICATIONS OF THIS CHOICE

What Does This Choice Mean for My Research?

Aspect	Implication
Ontology (nature of reality)	I recognize that construction projects have an objective reality (actual costs, actual durations) and a subjective reality (managers' decisions, risk perceptions). The artifact must operate in both dimensions.
Epistemology (how we know)	Knowledge is generated through the construction and evaluation of the artifact. Each iteration of the model produces new knowledge about the domain.

Methodology	I follow Hevner's 7 guidelines: build a relevant artifact, evaluate it rigorously, and communicate results to technical and managerial audiences.
Validation	The artifact is validated by its utility: predictive accuracy (MAPE < 12%), deviation reduction (>30%), and accessibility (cost < $15,000).

What Type of Contribution Do I Expect to Generate?

1. Functional artifact: A predictive digital twin model that construction companies can implement.

2. Reference architecture: A low-cost digital twin design adapted to the Peruvian context.

3. Empirical evidence: Quantitative results on the reduction of cost overruns and delays.

4. Methodological knowledge: A ranking of machine learning algorithms for the Peruvian construction context.

5. Practical guide: A document in Spanish explaining how to implement the model.

5. A TENSION I HAVE NOT YET RESOLVED

I want to be honest about a tension I have not yet fully resolved. Design science allows me to build an artifact, but I am concerned that, in my enthusiasm for building, I may neglect a deep understanding of the human context in which that artifact will operate.

The project managers I have spoken with have told me things like: "historical data does not reflect the unique conditions of each project" (Paredes & Quispe, 2024). This statement, although debatable, reflects a real distrust toward predictive models. How can I build an artifact that is technically excellent, but that professionals do not want to use?

This tension—between technical excellence and human acceptance—is something I am still processing. I do not have a definitive answer. My intuition tells me that the solution lies in participatory design: involving project managers in the development of the artifact from the early stages. But I do not know exactly how to do this in practice. This is a topic I hope to explore further in the upcoming sessions of the course.

6. REFERENCES

Flyvbjerg, B. (2024). How big things get done: The surprising factors behind every successful project. McKinsey Global Institute.

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. MIS Quarterly, 28(1), 75-105.

Paredes, J., & Quispe, R. (2024). Percepciones sobre la analítica predictiva en la mediana construcción limeña: Un estudio cualitativo. Revista Ingeniería de Construcción, 39(2), 145-162.

Simon, H. A. (1996). The sciences of the artificial (3rd ed.). MIT Press.e